

10 — Persona Card Schema

Status: DRAFT v0.1 (2026-06-11)

The single, common schema for ALL hierarchy levels. SO, ASM, RSM, NSM, Brand Manager, Marketing Manager, Business Analyst are all the SAME shape with different values.

Purpose

A Persona Card is the **vendor-curated, industry-generic blueprint for one role**. It is the secret sauce: it encodes CPG domain knowledge (hierarchy, what the role watches, how it disambiguates, what it asks) so a buyer only has to assign users to it. Cards are shipped and versioned by the vendor; buyers do not normally edit them (see 20-resolution.md).

Design rules

- **Common to all levels.** Differences are expressed as field VALUES, never new structure.
- **Semantic, not visual.** No layout/CSS here — see 14-section-grammar.schema.md .
- **Thin.** kpis , insight_types , questions[].intent , and catalog slots are *references* into 12 (metrics), 13 (insight rules), and 11 (catalog). The card names them; it does not define their logic or thresholds.
- **Deterministic-safe.** Nothing here produces a metric value or a score.

What makes one schema cover every level

Axis that varies by level	Expressed as	Leaf vs top example
Position in org	hierarchy.{tree,level,reports_to,manages}	SO manages: [] · NSM manages: [rsm]
Unit of accountability	scope.operating_grain + paths	SO outlet · NSM nation
Org tree	hierarchy.tree	sales vs marketing

Axis that varies by level	Expressed as	Leaf vs top example
What they watch / ask	list contents of kpis / insight_types / questions / actions	different members, same keys

Field table

Field	Type	Required	Notes
card_id	string (snake_case)	yes	stable, vendor-owned, unique
version	integer	yes	bumped on vendor upgrade
display_name	string	yes	human label
job_description	string	yes	plain-English job of this role. Sets the assistant's TOPIC scope: questions inside the job are handled, questions outside are declined. Does NOT control data access — that is RBAC.
hierarchy.tree	enum sales marketing analytics	yes	which org tree
hierarchy.level	integer	yes	rank within tree (higher = more senior)
hierarchy.reports_to	string null	yes	card_id of manager, null at top
hierarchy.manages	string[]	yes	card_id s managed; [] for a leaf
scope.operating_grain	string (dimension)	yes	unit the role owns (outlet , distributor , region , nation , brand ...)

Field	Type	Required	Notes
scope.drilldown_path	string[]	yes	ordered dims the role can go DOWN
scope.rollup_path	string[]	yes	ordered dims the role rolls UP into (compare/escalate); may be []
scope.default_period	enum TODAY MTD QTD YTD	yes	default time framing
kpis.primary	string[] (metric ids)	yes	headline KPIs; ids must exist in 12
kpis.secondary	string[] (metric ids)	no	supporting KPIs
preferences	PrefRef[]	yes	catalog overrides this role cares about
preferences[].preference	string (preference id)	yes	must exist in 11
preferences[].default	string	yes	must be one of the preference's options in 11
preferences[].ask_priority	enum high medium never	yes	drives cold-start selection
insight_types	string[] (rule ids)	yes	subset of registry 13 shown to this role
questions	Question[]	yes	curated Ask-Analyst / clarification bank
questions[].id	string	yes	stable
questions[].text	string	yes	display text (may contain {{tokens}})
questions[].intent	string	yes	pre-classified intent id
questions[].entities	object	no	known entities (metric/dimension/period)
questions[].cube_context	object	no	measures/dimensions/filters hint for chat fast-path
actions	Action[]	yes	role-appropriate recommended actions

Field	Type	Required	Notes
actions[].id	string	yes	stable
actions[].label	string	yes	display verb phrase
actions[].from_insight	string (rule id)	no	insight type this action maps from
cold_start.ask_preferences	string[] (preference ids)	yes	subset of preferences[].preference , all ask_priority: high
cold_start.max_questions	integer (1–3)	yes	hard cap on cold-start questions

JSON Schema

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://cpg-analytics/spec/persona-card.schema.json",
  "title": "PersonaCard",
  "type": "object",
  "additionalProperties": false,
  "required": ["card_id", "version", "display_name", "job_description", "hierarchy", "scope",
    "kpis", "preferences", "insight_types", "questions", "actions", "cold_start"],
  "properties": {
    "card_id": { "type": "string", "pattern": "^[a-z][a-z0-9_]*$" },
    "version": { "type": "integer", "minimum": 1 },
    "display_name": { "type": "string", "minLength": 1 },
    "job_description": { "type": "string", "minLength": 1 },

    "hierarchy": {
      "type": "object",
      "additionalProperties": false,
      "required": ["tree", "level", "reports_to", "manages"],
      "properties": {
        "tree": { "enum": ["sales", "marketing", "analytics"] },
        "level": { "type": "integer", "minimum": 1 },
        "reports_to": { "type": ["string", "null"], "pattern": "^[a-z][a-z0-9_]*$" },
        "manages": { "type": "array", "items": { "type": "string", "pattern": "^[a-z][a-z0-9_]*$" } }
      }
    },

    "scope": {
      "type": "object",
      "additionalProperties": false,
      "required": ["operating_grain", "drilldown_path", "rollup_path", "default_period"],
      "properties": {
        "operating_grain": { "type": "string" },
        "drilldown_path": { "type": "array", "items": { "type": "string" } },
        "rollup_path": { "type": "array", "items": { "type": "string" } },
        "default_period": { "enum": ["TODAY", "MTD", "QTD", "YTD"] }
      }
    },

    "kpis": {
      "type": "object",
      "additionalProperties": false,
      "required": ["primary"],

```

```
"properties": {
  "primary": { "type": "array", "minItems": 1, "items": { "type": "string" } },
  "secondary": { "type": "array", "items": { "type": "string" } }
},

"preferences": {
  "type": "array",
  "items": {
    "type": "object",
    "additionalProperties": false,
    "required": ["preference", "default", "ask_priority"],
    "properties": {
      "preference": { "type": "string" },
      "default": { "type": "string" },
      "ask_priority": { "enum": ["high", "medium", "never"] }
    }
  }
},

"insight_types": { "type": "array", "items": { "type": "string" } },

"questions": {
  "type": "array",
  "items": {
    "type": "object",
    "additionalProperties": false,
    "required": ["id", "text", "intent"],
    "properties": {
      "id": { "type": "string" },
      "text": { "type": "string" },
      "intent": { "type": "string" },
      "entities": { "type": "object" },
      "cube_context": { "type": "object" }
    }
  }
},

"actions": {
  "type": "array",
  "items": {
    "type": "object",
    "additionalProperties": false,
    "required": ["id", "label"],
    "properties": {
```

```

    "id": { "type": "string" },
    "label": { "type": "string" },
    "from_insight": { "type": "string" }
  }
},
{
  "cold_start": {
    "type": "object",
    "additionalProperties": false,
    "required": ["ask_preferences", "max_questions"],
    "properties": {
      "ask_preferences": { "type": "array", "items": { "type": "string" } },
      "max_questions": { "type": "integer", "minimum": 1, "maximum": 3 }
    }
  }
}
}

```

Validation rules (beyond JSON Schema — enforce in CI)

1. Every `id` in `kpis.primary` / `kpis.secondary` MUST exist in the metric registry (12).
2. Every `id` in `insight_types` MUST exist in the insight-rule registry (13).
3. Every `preferences[].preference` MUST exist in the catalog (11); its `default` MUST be one of that preference's `options` .
4. Every `cold_start.ask_preferences` entry MUST appear in `preferences` with `ask_priority: high` .
5. `len(cold_start.ask_preferences) <= cold_start.max_questions` .
6. `hierarchy.reports_to` and every `hierarchy.manages id` MUST resolve to an existing `card_id` in the SAME tree (referential integrity across the persona set).
7. `scope.operating_grain` and all path `dims` MUST be valid Cube.js dimensions.
8. `job_description` sets TOPIC scope only. It MUST NOT be used to grant or restrict data access. RBAC is the only access control (see 20-resolution.md).
9. Every rule `id` in `insight_types` SHOULD list the card's `scope.operating_grain` in its `applies_to.operating_grain` (or omit `applies_to`). Keeps shown insights grain-appropriate.
10. Every rule `id` in `insight_types` SHOULD have a matching card action — an `actions[]` entry whose `from_insight` equals that rule `id` — so every shown insight is actionable.

Versioning

- Additive field changes → minor; the schema `version` in this doc bumps.
- Breaking changes (renamed/removed field, tightened enum) → major; authored cards must declare the schema version they target and be migrated.
- A card's own `version` is independent — it tracks vendor content upgrades to that role.

Worked example

See `examples/asm.card.yaml` (Area Sales Manager) and `examples/nsm.card.yaml` (National Sales Manager) — identical shape, different values.